Sound Classification using the Broad Sound Taxonomy

Frederic Vaquer

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1 Data

For this assignment, I curated a subset of classes from the BSD10k dataset. I selected 15 of the original 23 classes based on personal interest and the distinct characteristics of the sounds. This subset comprises a total of 7,205 audio files along with their respective embeddings, as illustrated in Figure 1. For all experiments, the training and test sets were precomputed according to the dataset's official metadata.

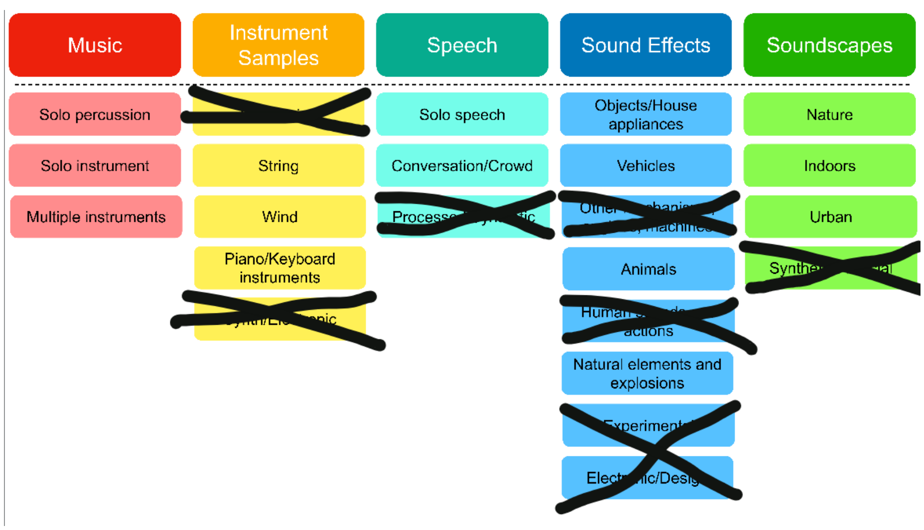


Figure 1: BSD10k diagram and kept classes

2 Experiments

Regarding the experiments, I explored two distinct approaches using the curated subset. First, I performed classification specifically on the 'top' classes identified in Figure 1. Second, I expanded the task to include the entire subset. This was intended to determine whether increasing the complexity of the training task improves performance on the specific classes of interest—an approach inspired by multitask learning, where models are trained on broader objectives to enhance performance on primary tasks.

Furthermore, I utilized the FreeSound API to retrieve audio files based on specific queries. This allowed me to evaluate how the top-performing classifier categorized these new sounds and to assess whether the predictions aligned with both human perception and the metadata provided by FreeSound. Finally, I experimented with CLAP embeddings and text-based queries to investigate how varying the precision and specificity of queries influences the relevance of the retrieved audio. I conducted three specific experiments in this domain: basic query usage, the impact of query specificity, and the precision of the results based on whether the expected label appeared within the five nearest neighbor classes.

3 Results

3.1 Classification analysis

Comparing the overall accuracy of the different approaches, as shown in Figure 2, we observe that performance trends remain consistent across embedding types: CLAP consistently yields the highest accuracy, while MFCC performs the worst.

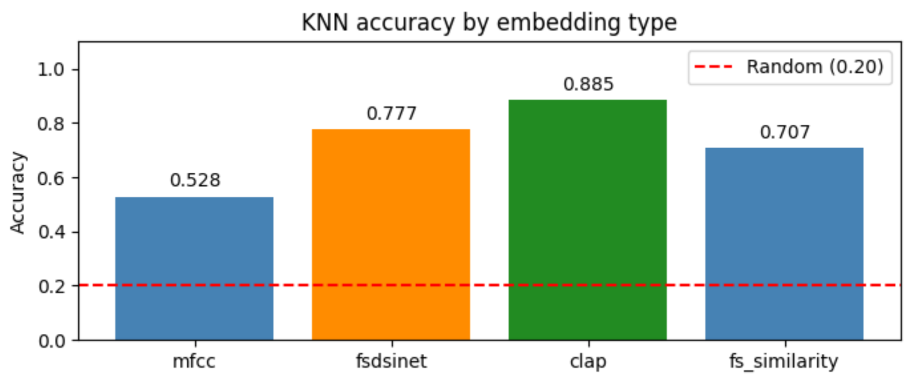
A detailed analysis of misclassifications is provided in the appendix, where the density of the confusion matrices warrants dedicated figures. Figure 3 displays the confusion matrices for the full subset classifiers using k-NN with different embeddings. Additionally, Figure 4 compares the performance of classifiers trained specifically on the top-level classes versus those trained on the full subset. Notably, the CLAP confusion matrix aligns closely with the results reported in the reference literature.

Regarding the subset classifiers, the MFCC model exhibits significant off-diagonal dispersion, suggesting that its representational capacity is insufficient for this broad sound taxonomy. Conversely, other embeddings demonstrate superior performance, with CLAP emerging as the clear leader. While the misclassification patterns are generally similar across the more robust embeddings, the most prominent errors include:

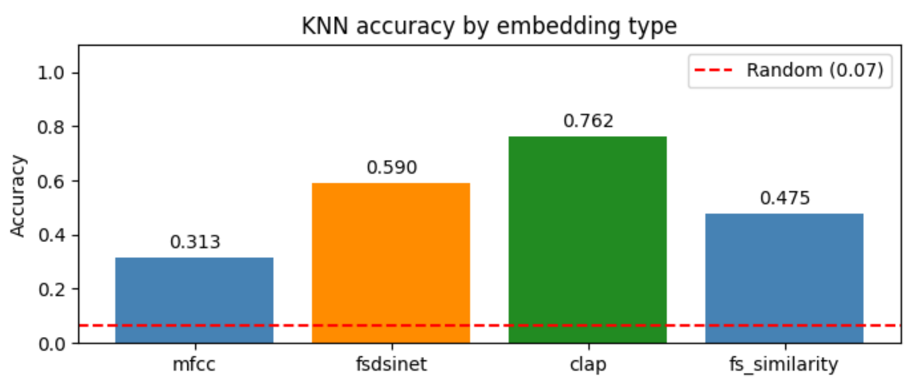
- **fx-a vs. ss-n**: This confusion is understandable, as the categories for animal sound effects and natural soundscapes often contain overlapping spectral characteristics.

- **fx-a vs. fx-n**: This highlights a semantic ambiguity; the boundary between natural elements and animal sound effects is often subtle, even to human listeners.

- **sp-c vs. ss-u**: This indicates difficulty in separating conversation or crowd sounds from urban soundscapes, as contexts like a discussion in a park contain elements of both classes.



(a) Accuracies for top genre classification



(b) Accuracies for sub genre classification

Figure 2: Accuracies for both approaches

Regarding the utility of training on sub-classes to improve top-level performance, the results show no clear advantage, as overall classification accuracy actually tends to decline. However, modest improvements were observed in the *Speech* and *Soundscape* categories. For *Speech*, this likely occurs because the high frequency of solo speakers (761 vs 56 conversations/crowd) in the training set causes models trained only on top-level classes to equate *Speech* strictly with solo voices, misclassifying multi-person interactions as *Soundscapes* instead. Similarly, the gains in *Soundscapes* appear to stem from a more refined taxonomy, where clearer distinctions between nature sounds and instrument samples reduced label ambiguity, thereby improving the model's ability to isolate the *Soundscape* characteristics within the broader sub-classification task.

However, the this improvements are worse as the embedding type performed better (i.e., for MFCC the improvement in these classes is greater than for CLAP embeddings).

When evaluating misclassifications through manual inspection, it becomes clear that some errors are inevitable even for human listeners. For instance, a recording of multiple speakers was misclassified as *Solo Speaker* because the voices did not overlap, leading the model to treat the audio as a single-speaker event. In another case, an audio sample labeled *FX-Animal* was predicted as *Soundscape Nature*; upon closer inspection, the presence of subtle background noises made this an understandable misclassification even by human standards. Finally, electronic music samples labeled *Multiple Instruments* were often categorized as *Solo Percussion* or *Solo Instrument*, depending on whether the texture was rhythmic or melodic. Maybe, like for Instrument Samples, there should be a Music electronic subclass, allowing for such samples to have their own label.

3.2 Classification using FreeSound API

The results exhibit a noticeable bias toward the *Sound Effects* class, which is expected given its prevalence (over 2,000 samples) in the training data. While dog barks were correctly identified, the rain sample was classified as an *Explosion*—a counter-intuitive result having a *Soundscape Nature* class, yet understandable when listening to the sample and considering the mentioned bias toward the *Sound Effects* class. Other classifications reveal interesting nuances: an acoustic guitar chord was categorized under *Sound Effects*—plausible but less precise than *Instrument Samples*—and a cat's meow was labeled *Soundscape Nature*, likely due to the ambient silence surrounding the sound.

For out-of-domain samples, the model struggled with synthetic inputs: a sine wave, Geiger counter, white noise, and digital glitch were all erroneously classified as *FX-Animal*. While the rhythmic pulse of a Geiger counter might mimic animal patterns, these errors highlight the model's inability to process synthetic audio after such categories were excluded from the training set. Notably, these misclassifications consistently yielded the lowest k-NN voting strength, indicating high model uncertainty. Conversely, the hydraulic piston was reasonably classified as *FX-Object*, while the telephone dial tone was incorrectly assigned to *FX-Natural Element*, likely a limitation imposed by the subset's restricted taxonomy, which lacks specific categories for electronic notifications.

3.3 CLAP text queries

Regarding query-based retrieval, general prompts yielded high relevance, though specific failures occurred: *crowd applauding* and *baby crying* presented the greatest challenges. Notably, the *baby crying* query retrieved sounds of distressed animals, highlighting a semantic overlap in the latent space.

Prompt sensitivity tests demonstrated that the model effectively captures nuances in specificity. For instance, transitioning from *rain* to *heavy rain hitting a metal roof* consistently yielded increasingly tailored samples. Similarly, guitar-related queries successfully navigated from general *Solo Instruments Music* categories to more granular *String Instrument Samples* classes. However, performance waned with the *person speaking quietly indoors* query; while *voice* and *speech* were retrieved accurately, the addition of the "indoors" constraint shifted the model's focus toward *Soundscape* characteristics, likely due to the dataset's limited representation of indoor speech environments.

Finally, as shown in Table **??**, precision is highly dependent on query complexity and domain coverage. For categories such as *crowd applauding* or *baby crying*, the model struggled to form a coherent cluster, likely because the necessary *Human Sounds* classes were absent from the subset. This underscores a limitation in the dataset's coverage, where specialized queries frequently encounter "out-of-domain" gaps that force the model to map inputs to the closest, albeit semantically distant, neighbors.

Query	**Expected Class**	**P@5**
rain falling on leaves	fx-n	1.00
dog barking	fx-a	0.80
crowd applauding	ss-u	0.60
electric guitar riff	m-si	0.80
car engine starting	fx-v	0.60
baby crying	sp-s	0.00
bird singing in a forest	ss-n	0.60

Table 1: Performance evaluation of query-based retrieval (P@5) using CLAP embeddings.

4 Conclusion

In conclusion, CLAP outperformed the other embeddings, with misclassifications often stemming from genuine semantic ambiguities—such as the overlap between natural sounds and animal effects—that can be challenging even for human listeners. While CLAP is a powerful tool for text-to-audio retrieval and shows sensitivity to query specificity, it occasionally encounters limitations due to the database's size and structure. Some highly specific queries map to sounds that are close in the latent space but contextually unrelated, highlighting the influence of the training data on retrieval performance. Ultimately, while CLAP effectively bridges natural language and audio, these results suggest that further refinements in taxonomy and data coverage could help mitigate these mapping errors.

A Supplementary Figures

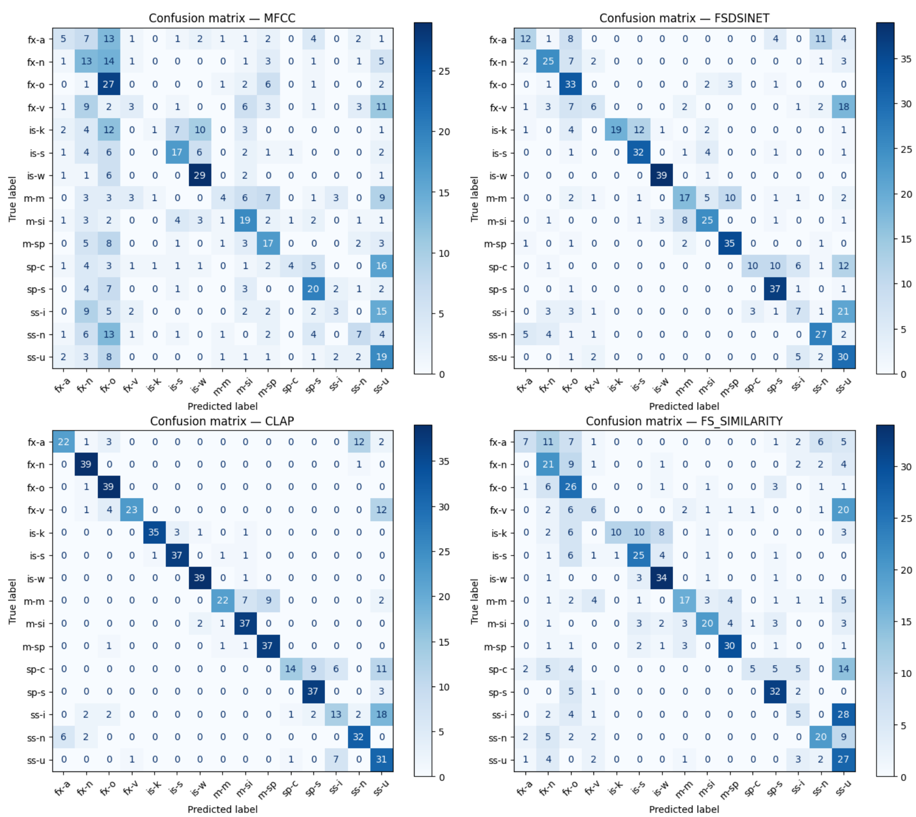
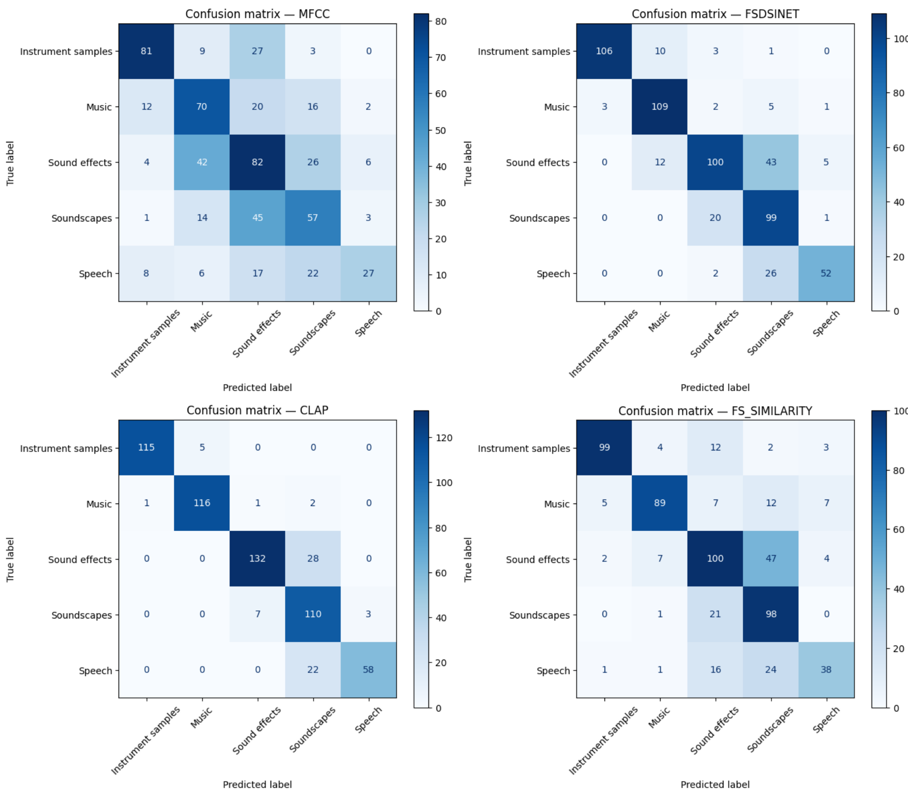
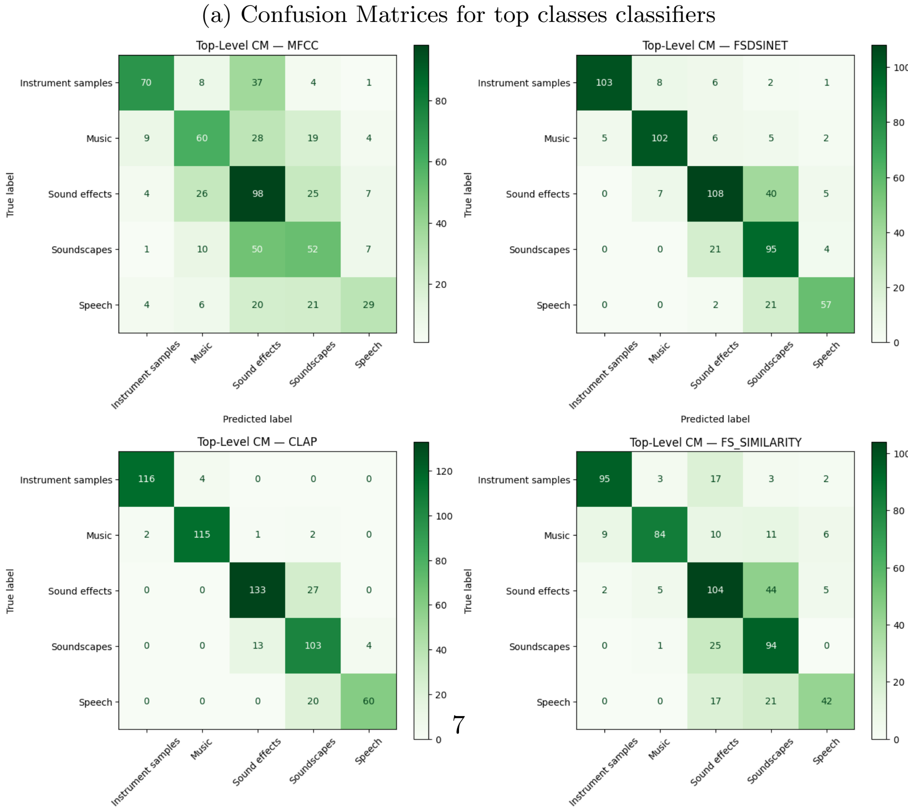


Figure 3: Confusion matrices over the preserved classes



(a) Confusion Matrices for top classes classifiers



(b) Confusion Matrices for sub classes classifiers mapped to top genre

Figure 4: Confusion matrices over the top classes